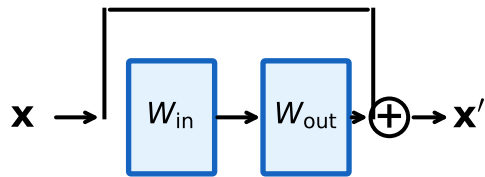


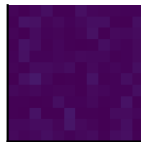
(a) The Object: Residual Branch Product



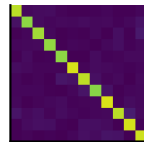
$$M = W_{\text{out}} W_{\text{in}}$$

$$s(M) = \frac{|\text{tr}(M)|}{\|M\|_F}$$

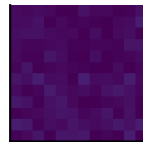
(b) The Signal: Diagonal Emerges Only in Trained Residual



Random Init
(2% acc)



Trained ResNet
(100% acc)



Trained PlainNet
(3% acc)

(c) The Application: Lineage Verification

